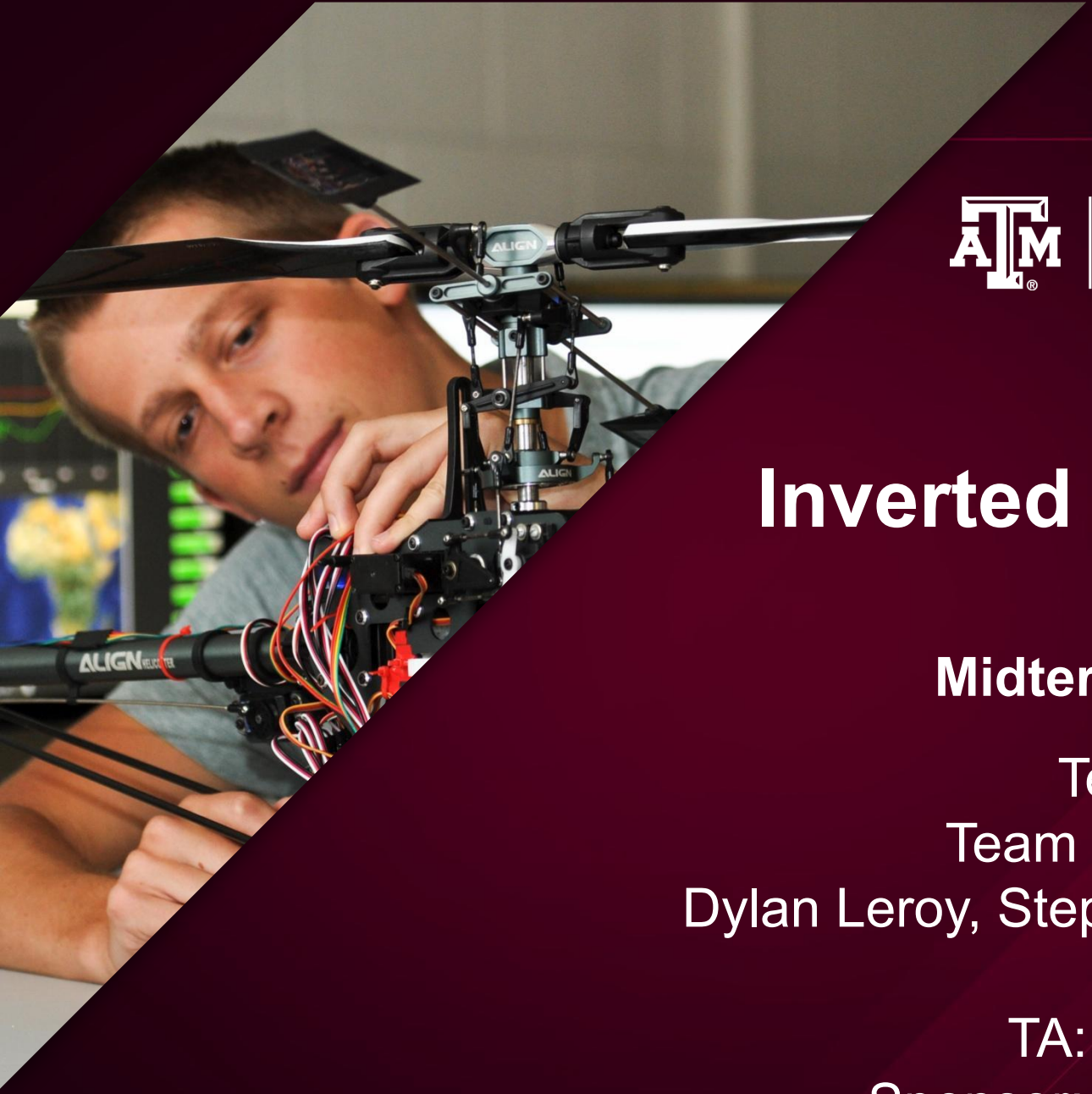


Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 403
Inverted Pendulum Robot 3
Midterm Presentation

Team Number: 55
Team Lead: Sean Wall
Dylan Leroy, Stephen Darwin and Arian Golkarfard
TA: Swarnabha Roy
Sponsor: Aniruddha Datta

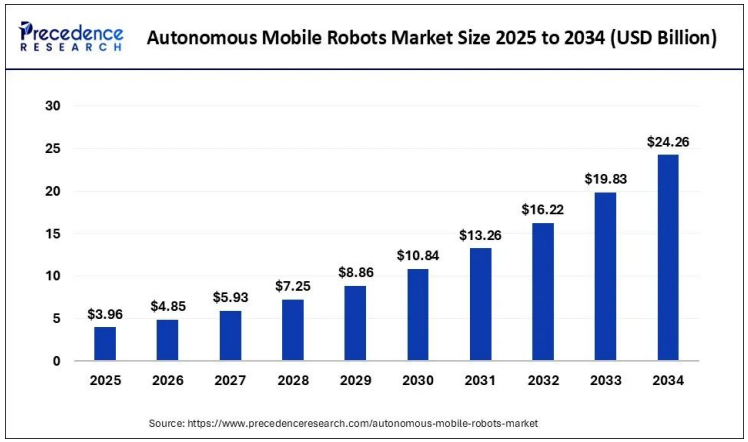
Project Summary

Problem statement:

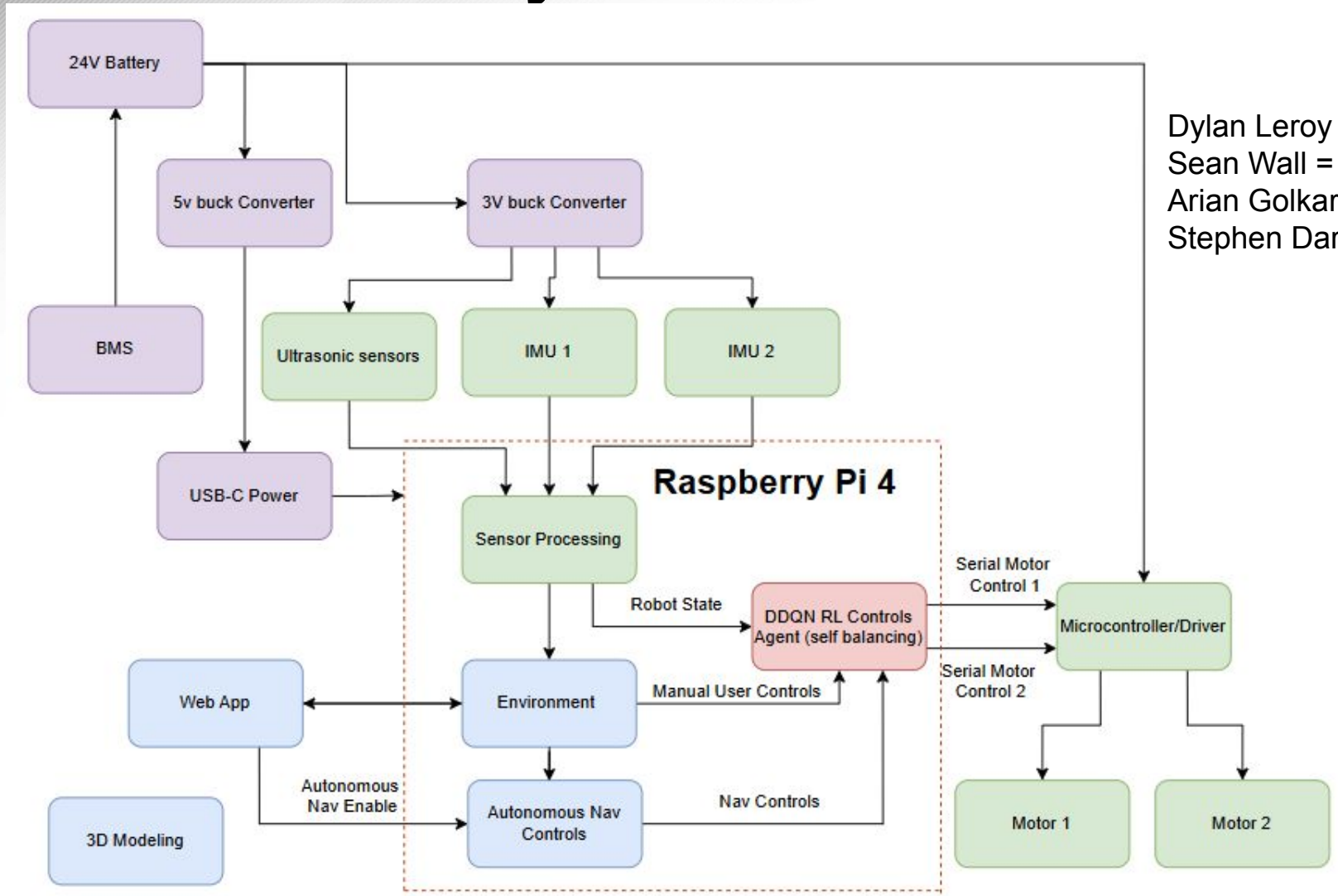
- Current mobile robot applications are plagued by rigid and inflexible control systems with limited ability to adapt to dynamic environments. Specifically, warehouse environments would benefit from reliable and robust methods for transporting packages from one location to another.

Our Solution:

- We will implement a reinforcement learning algorithm as the controls system of our two wheeled balancing robot. This will be trained extensively in both simulation, and on real hardware.
- This will result in a robust controls system easily capable of adapting to various loads and operating conditions.



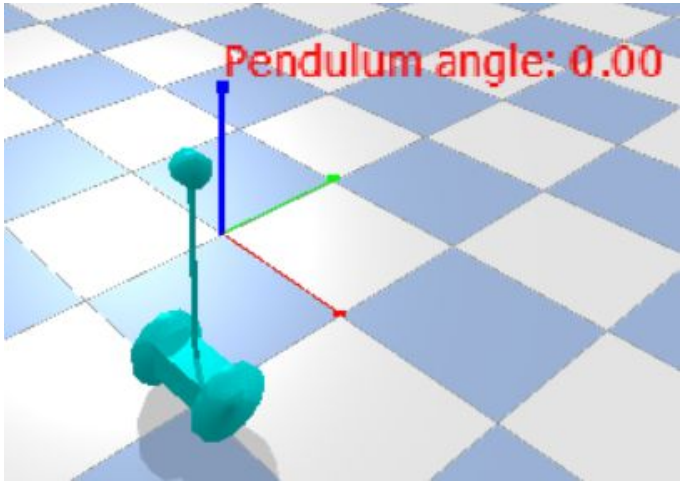
System Overview



Dylan Leroy = Red
Sean Wall = Blue
Arian Golkarfard = Green
Stephen Darwin = Purple

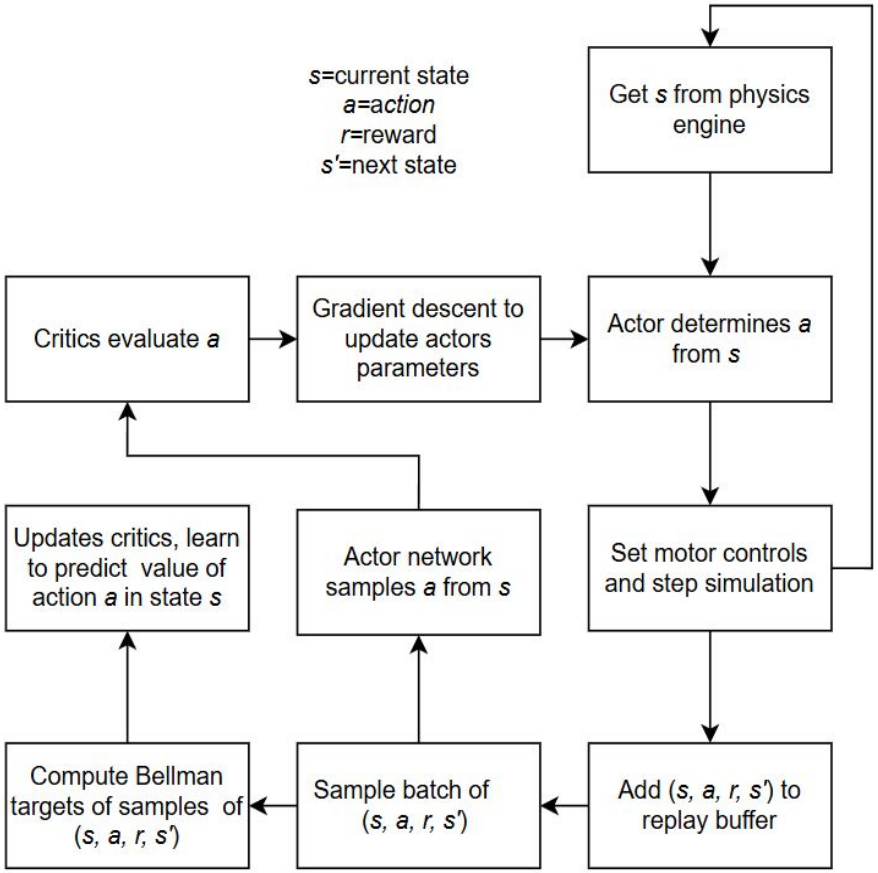
RL Controls System – Dylan Leroy

- Motor control for balancing and simple movement handled by a neural network trained by Soft Actor-Critic (SAC) reinforcement learning algorithm
- Use Stable-Baselines3's SAC library alongside PyBullet to train agent in simulation
- Will learn to balance, and take commands to move back and forth, as well as turn
- Will generate values from -1 to 1 for each motor which will map to the serial values for our motor controller

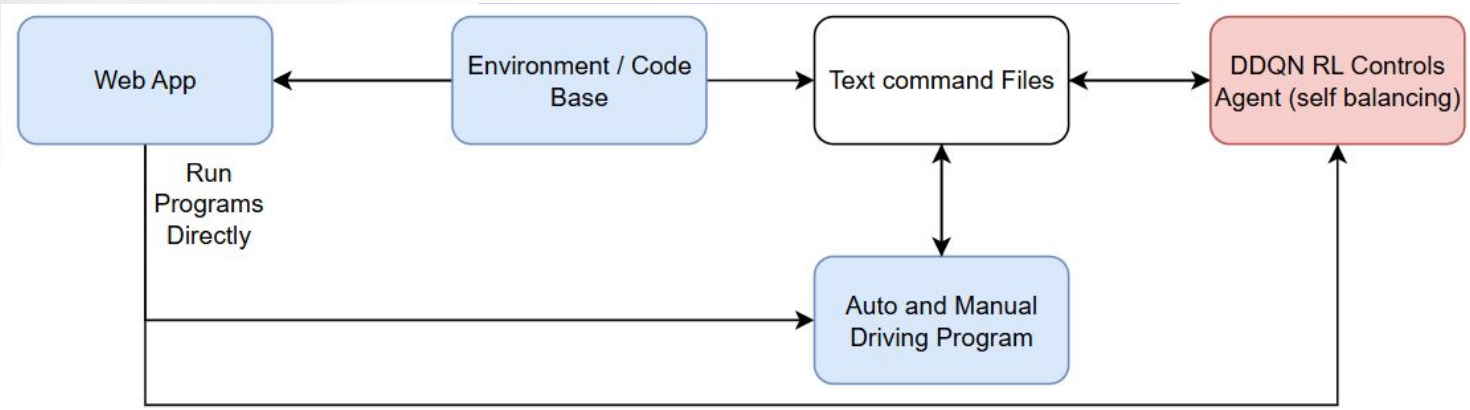


Soft Actor-Critic (SAC)

	DDQN	SAC
PROS	Verified to be a top performing RL training algorithm for a two wheeled balancing robot.	Continuous output, better for smooth control, robust algorithm with efficient data use and entropy variables to ensure exploration.
CONS	Discrete outputs only, choppy movement.	No verification of performance on this specific application.



Web App/Database - Sean Wall



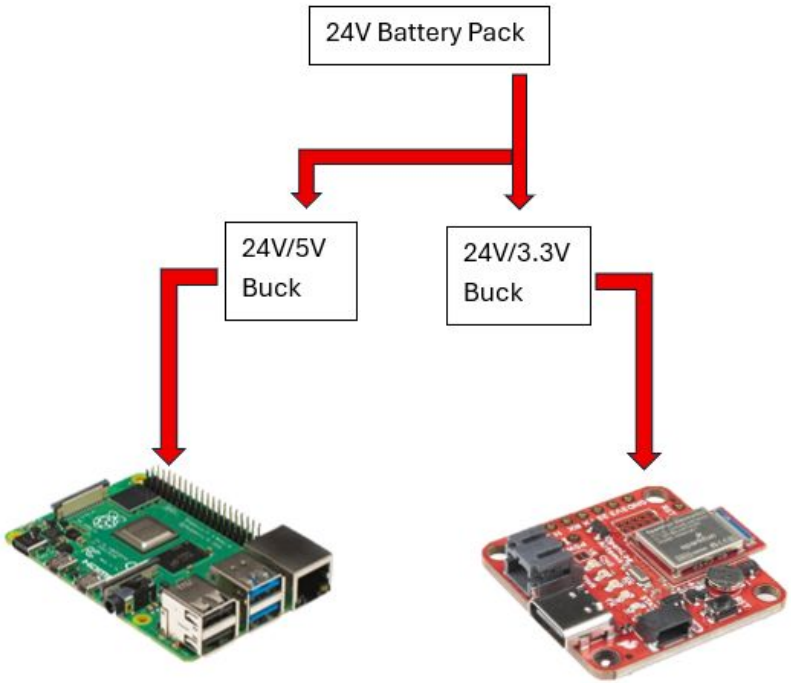
- This subsystem is the user control interface for the whole robot.
- This subsystem is being designed on the **Raspberry Pi 4** using a flask environment to allow a python based web app to be used from a phone
- This subsystem will collect and display data from the controls and sensor subsystems, alongside running and stopping the other programs.
- In the background, there will also be development of the 3D model.

Mobile App Tradeoffs

	Flutter	Flask environment	X- Code
PROS	Has cross platform UI (Ios, Android, Windows) Fast Prototyping.	Lightweight and easily modifiable Full control over structure of website, libraries and integration Runs Locally.	Native apple tools good hardware.
CONS	Not ideal for real time hardware May have some performance issues.	Lacks security and is mostly for local use. Uses Python.	Not compatible with android or windows Would have some Issues being able to test due to this.

Power System – Stephen Darwin

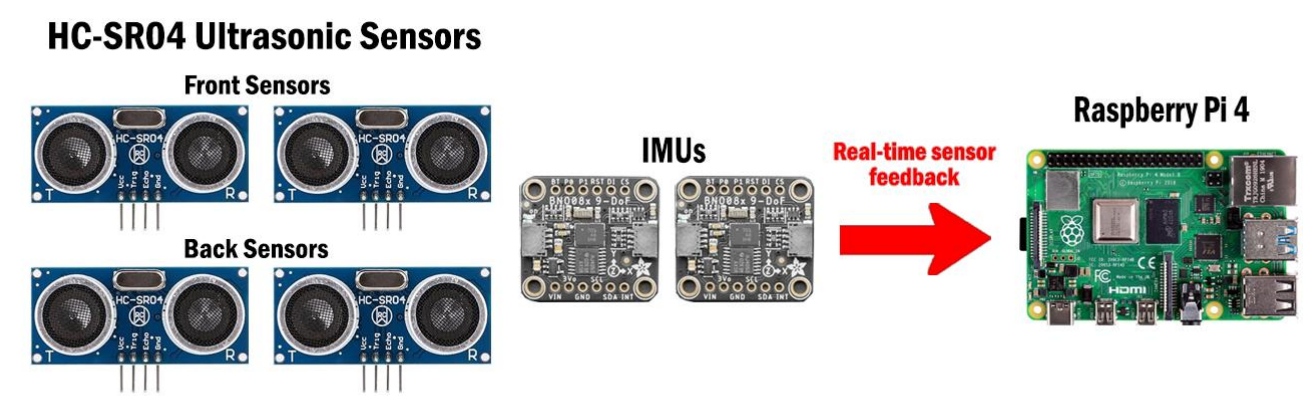
- **Purpose:** Provide stable/usable DC voltages to Raspberry Pi 4 (5V/3A) and IMU (~3.3V/3mA)
- **Process/Design:**
  - Using TPS54560B Buck Regulator IC
  - Buck converter schematic design and simulation using Altium and LTSpice
  - PCB design and component BOM generated using Altium Designer in conjunction with relevant TI component datasheet
- Two **separate** buck converters
  - 24V to 5V for Raspberry Pi
  - 24V to 3.3V for IMU/other sensors
- Both bucks will support up to 5A of load current



Buck Regulator IC Tradeoffs

	SiC431	TPS54560B	LM2596
PROS	Newer IC with highly customizable output voltage and good load capacity.	Input/output voltages and load capabilities exceed spec comfortably. Available to simulate on WEBENCH.	Low complexity, available to simulate on WEBENCH, highly customizable input and output voltage.
CONS	High complexity, very little headroom for input overvoltage. No reliable existing simulation resources.	Intermediate complexity, requires numerous external connections. Older IC, not the “latest and greatest”.	Max load current leaves NO room for current spikes at startup. Would require unusually large inductor (47uH) to smooth output ripple.

Sensor Feedback – Arian Golkarfard



- Uses 4x **HC-SR04 Ultrasonic sensors** (2 front, 2 back).
- Provides distance measurements to detect obstacles for real-time positional feedback into navigation and balancing algorithms.
- Works alongside IMU sensors to improve positional awareness.
- Integrated with Raspberry Pi 4 for signal processing and control.
- The sensors measure accurately within a range of about 2 to 400 cm with millimeter-level resolution and low latency.
- One key constraint is signal interference with multiple sensors working at once.

Sensor Tradeoffs

	LiDAR Sensors	Ultrasonic Sensors
PROS	Offer high precision distance measurements over long range with fast scanning capability.	Provide reliable short to medium range distance measurements at low cost with simple integration.
CONS	Expensive and power-hungry, making them less suitable for small low-cost robots.	Limited range and resolution, and potential interference between multiple sensors.

Motor Controller – Arian Golkarfard

- **Sabertooth 2x12:** Dual-channel motor control for up two DC motors at 12 A
- Interfaces directly with the Raspberry Pi 4 using simplified serial.
- Supports mixed and independent motor modes for differential drive.
- Integrates current limiting, thermal protection, and regenerative braking for safety and reliability.
- Reduces design complexity by using a pre-built, tested module instead of a custom H-Bridge design.

Project Execution & Validation

	10/1 - today	10/6	10/14	11/3	11/17
RL Controls System (Dylan)	Describe robot in URDF and setup PyBullet environment	Integrate SAC agent with simulation environment	Train agent to balance	Train agent to move	Validate with navigation tests in simulation
Web App/Database (Sean)	Was able to remotely deliver commands and activate programs	Send signals out the GPIO of the Raspberry Pi	Continue Developing the website to interact with the real programs	Develop the User Interface to make it easier to use with more functionality	Validate the usage and create setup for Final Presentation
Power (Stephen)	Completing buck converter schematics, PCB files.	Finish simulating and validating PCB components and connectivity. Order components and boards.	Solder/build PCBs.	Continue validating PCBs.	Test completed PCBs.
Sensor Feedback & Microcontroller (Arian)	Research on ultrasonic sensing and IMU integration, and select desired microcomponents.	Build initial sensor interface on Raspberry Pi and order the parts and sensors.	Integrate four ultrasonic sensors, validate IMU accuracy, and test microcontroller.	Run subsystem tests with combined ultrasonic + IMU feedback	Validate subsystem with control tests, and confirm obstacle detection.